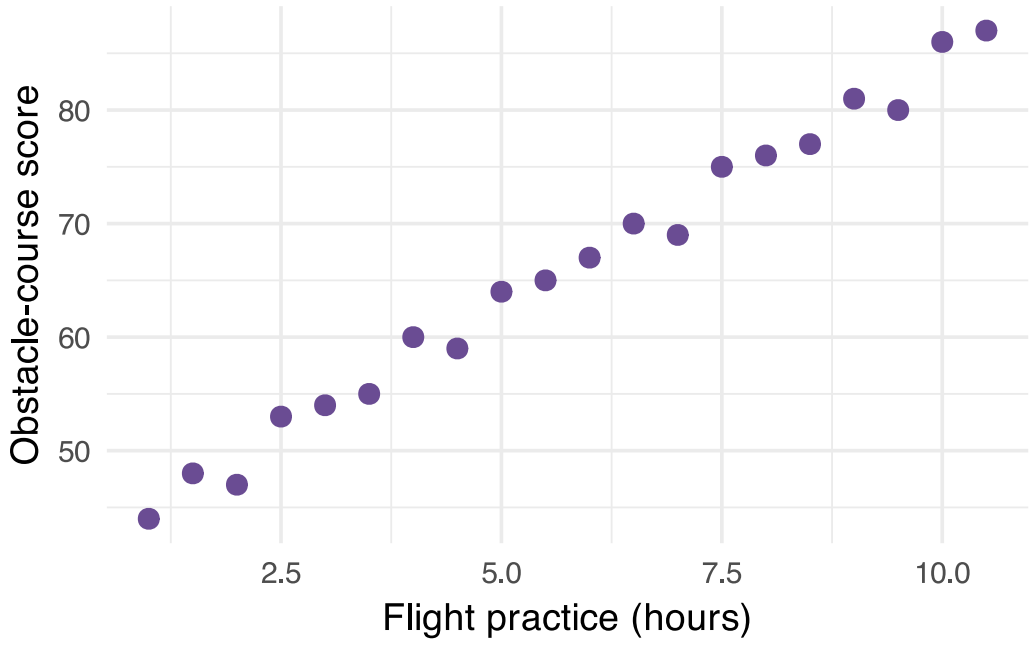
Example 1

- Rainbow Dash wants to know whether more flight practice tends to go with a higher score on the Wonderbolts obstacle course.
- Our observational unit is a pony. For each pony we have:
  - `practice_hours`: hours spent practicing
  - `course_score`: score on the course, from 0 to 100

```
flight <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA4173Fa
26/refs/heads/main/lectures/data/flight_practice_score.csv")
head(flight, n = 3)
```

```
# A tibble: 3 × 3
  pony_id practice_hours course_score
  <chr>            <dbl>        <dbl>
1 P01                1             44
2 P02                1.5           48
3 P03                2             47
```

- Let's examine the scatterplot,



- Is there anything unusual?
- What kind of relationship is this?

```
flight |> correlation()
```

```
# A tibble: 1 × 5
  var1         var2           correlation test_statistic p_value
  <chr>        <chr>                <dbl>          <dbl> <chr>
1 course_score practice_hours       0.995           40.7 < 0.001
```

- What is the correlation?

```
m1 <- glm(course_score ~ practice_hours, data = flight)
m1 |> tidy(conf.int = TRUE)
```

```
# A tibble: 2 × 7
  term           estimate std.error statistic  p.value conf.low conf.high
  <chr>             <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)       40.4      0.698      57.9 6.58e-22    39.1      41.8
2 practice_hours     4.42     0.109      40.7 3.57e-19     4.21      4.63
```

- What is the resulting regression model?

- What is the interpretation for the slope?

- What is the interpretation for the y-intercept?

- What is the confidence interval for β_1?

- Is x a significant predictor of y?

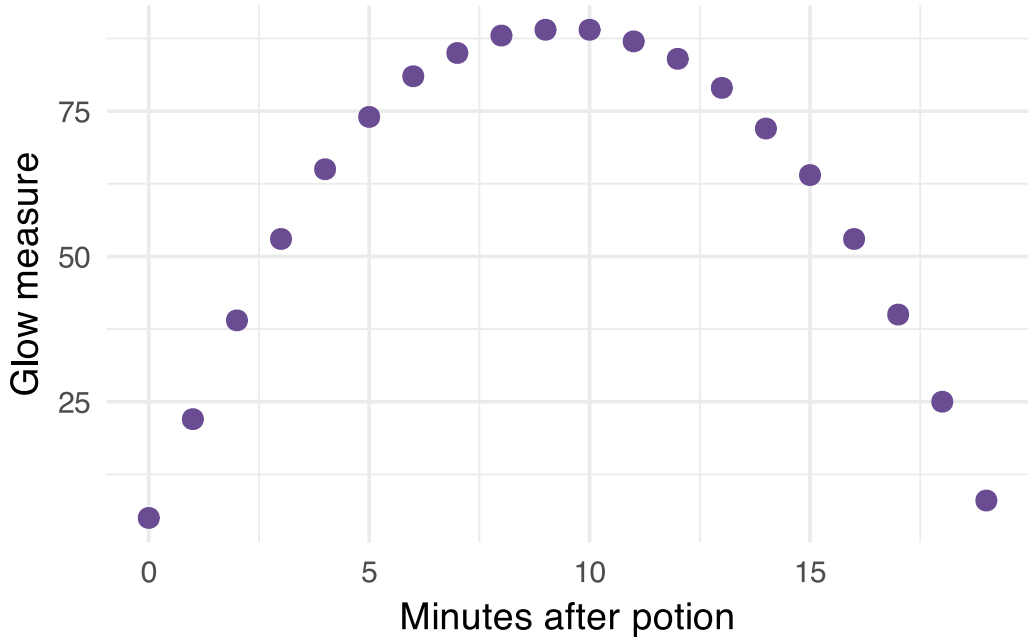
Example 2

- Twilight measures a spell's glow after a potion is cast. The glow rises, peaks, and fades.

```
curve_data <- read_csv("https://raw.githubusercontent.com/samanthaseals/STA4173Fa26/refs/heads/main/lectures/data/potion_response.csv", show_col_types = FALSE)
head(curve_data, n = 3)
```

```
# A tibble: 3 × 3
  trial_id minutes_after_potion glow_measure
  <chr>                   <dbl>        <dbl>
1 T01                         0            5
2 T02                         1           22
3 T03                         2           39
```

```
ggplot(curve_data, aes(minutes_after_potion, glow_measure)) +
  geom_point(size = 3, color = "#6A4C93") +
  labs(x = "Minutes after potion", y = "Glow measure") +
  theme_minimal(base_size = 14)
```



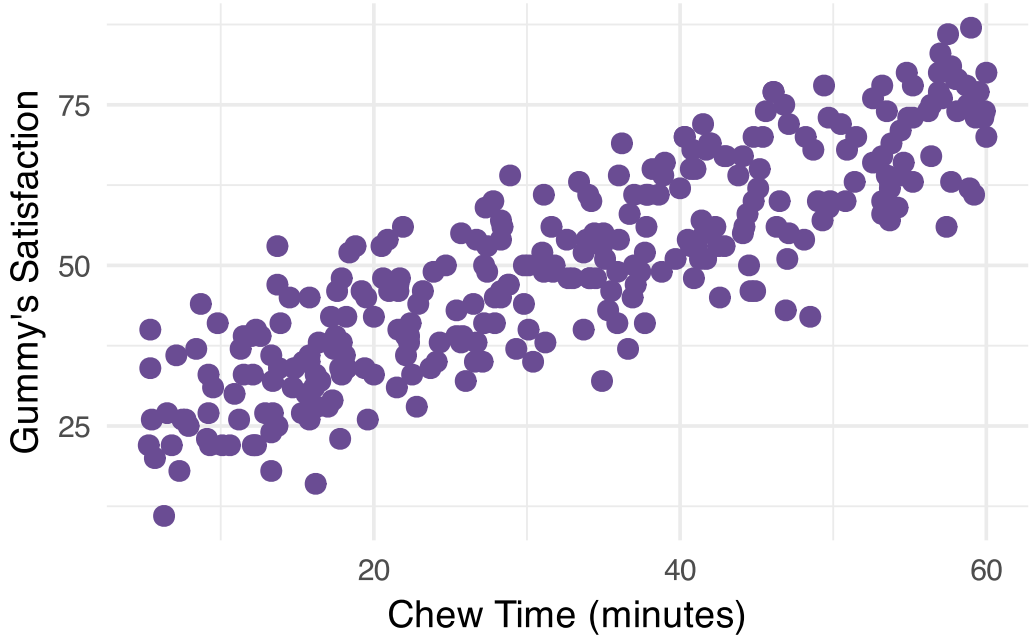
- Would one straight line describe this pattern well?

Example 3

- In Ponyville, Pinkie Pie is curious about how Gummy's snack satisfaction (0 to 100) relates to the duration of chew time (in seconds) he spends on crunchy treats. She suspects that Gummy enjoys snacks more the longer he chews them.
- Pinkie Pie records both the chew time (*chew_time*) and Gummy's satisfaction level (*satisfaction*) of the last 300 snacks given to Gummy (**gummy_data**).

```
# A tibble: 3 × 4
  chew_time snack_crunch nap_time satisfaction
      <dbl>        <dbl>    <dbl>        <dbl>
1      38.8         67.6     27.9           49
2      13.4         63       16.5           32
3       8.7         53.4     16             44
```

- Examining the scatterplot,



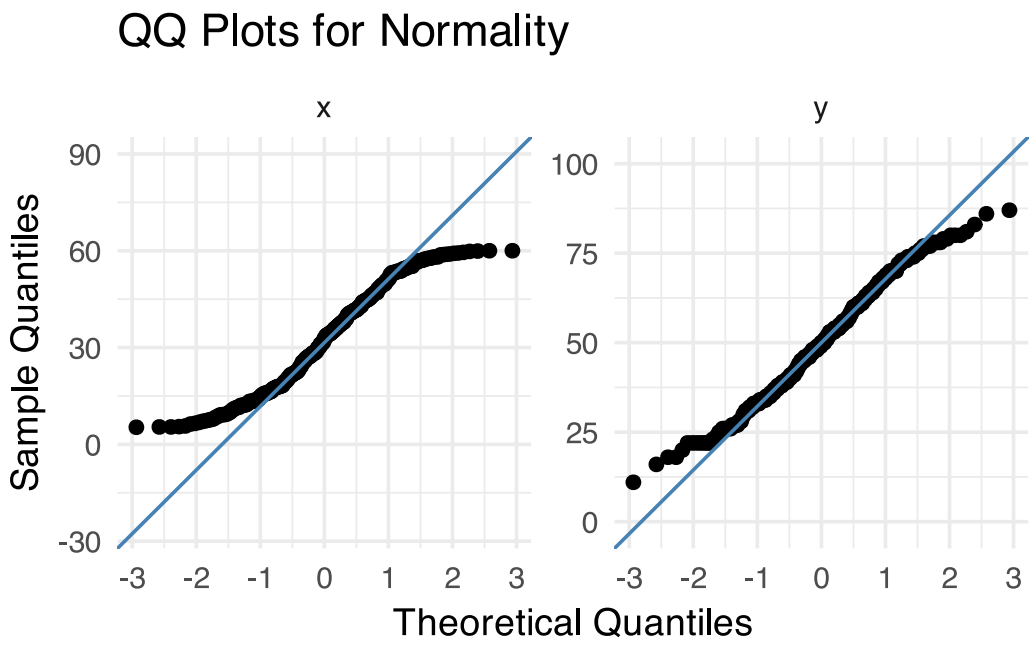
- Examining the correlation,

```
gummy_data |> correlation(x = chew_time, y = satisfaction)
```

```
# A tibble: 1 × 5
  var1      var2         correlation test_statistic p_value
  <chr>     <chr>              <dbl>          <dbl> <chr>
1 chew_time satisfaction       0.864           29.6 < 0.001
```

- Is this correlation significantly different from 0?

```
gummy_data |> correlation_qq(x = chew_time, y = satisfaction)
```



- Do we meet the normality assumption?

```r
gummy_data %>% correlation(x = chew_time, y = satisfaction, method = "spearman")
```

```
# A tibble: 1 × 5
  var1      var2         correlation test_statistic p_value
  <chr>     <chr>              <dbl>          <dbl> <chr>
1 chew_time satisfaction       0.868        594784. < 0.001
```

- Is the correlation is significantly different from 0?

```r
m3 <- glm(satisfaction ~ chew_time, data = gummy_data)
m3 |> tidy(conf.int = TRUE)
```

```
# A tibble: 2 × 7
  term        estimate std.error statistic  p.value conf.low conf.high
  <chr>          <dbl>     <dbl>     <dbl>    <dbl>    <dbl>     <dbl>
1 (Intercept)   21.1      1.09        19.4 9.50e-55   19.0      23.2
2 chew_time      0.894    0.0302      29.6 1.33e-90    0.834     0.953
```

- What is the resulting regression model?

- What is the interpretation for the slope?

- What is the interpretation for the y-intercept?

- What is the confidence interval for β_1?

- Is x a significant predictor of y?